METHYLPHENIDATE

Controlled drug

Monitor growth, BP, HR

CHEMICAL NAME Methyl (RS*,RS*)-2-phenyl-2-(piperidin-2-yl)acetate (hydrochloride)
CLASS CNS stimulant (dopamine/noradrenaline reuptake inhibitor)
TRADE NAMES Ritalin · Concerta XL · Medikinet · Equasym XL

THERAPEUTIC USES / INDICATIONS

- Attention deficit hyperactivity disorder (ADHD) in children ≥6 yr and adults, as part of a comprehensive programme.
- Narcolepsy.

ADMINISTRATION

- **PO** IR 2–3 times daily (last dose before 4 pm) or MR once daily in the morning (swallow whole).
- Drug holidays may be considered to assess need/growth.

DOSAGES

- **Children ≥6 yr (IR):** 5 mg once or twice daily, increase weekly by 5–10 mg; max 60 mg/day (90 mg specialist).
- **Concerta XL:** 18 mg once daily, up to 54 mg (adults 108 mg).
- **Adults:** 5 mg two or three times daily, max 100 mg/day.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Duration:** IR 3–5 h; MR 8–12 h.
- De-esterified in liver (carboxylesterase) — not CYP.
- Excreted in urine. **t½:** 2–3 h.

PHARMACODYNAMICS

- Blocks **dopamine and noradrenaline transporters** → ↑ catecholamines in prefrontal cortex → improved attention, impulse control, ↓ hyperactivity.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Decreased appetite, weight loss, insomnia.
- Headache, abdominal pain.
- Tachycardia, ↑ BP; irritability, anxiety.

ADVERSE EFFECTS (SERIOUS)

- Growth suppression in children (height, weight).
- Cardiovascular events in structural heart disease; arrhythmias.
- Psychosis, mania, tics; misuse/diversion.
- Priapism (rare); seizures.

CONTRAINDICATIONS & CAUTIONS

- Structural cardiac abnormalities, severe hypertension, arrhythmias, cerebrovascular disorders.
- Glaucoma; phaeochromocytoma; hyperthyroidism.
- Severe depression, anorexia nervosa, psychosis, bipolar (uncontrolled).
- MAOIs within 14 days.

KEY INTERACTIONS

- **MAOIs:** hypertensive crisis.
- **Antihypertensives:** ↓ effect.
- **Coumarins, phenytoin, SSRIs, TCAs:** ↑ levels.

NURSING MANAGEMENT

- Baseline cardiovascular history, BP, HR, height, weight.
- Record height, weight (centile chart), BP, HR every 6 months (and dose changes).
- Monitor sleep, appetite, mood, tics.
- Secure storage; controlled-drug procedures.

PATIENT EDUCATION

- Give in the morning (last IR dose before 4 pm) to protect sleep.
- Give with or after breakfast to reduce appetite loss.
- Report palpitations, chest pain, fainting, mood changes, tics.
- Keep medication locked away; do not share.

KEY POINTS

- CNS stimulant for ADHD.
- Appetite loss, insomnia, growth effects.
- Start 5 mg once–twice daily.
- Monitor BP, HR, height, weight.